BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2017

PHYSICS

Past Paper 2017 • Paper Code: A

Retyped faithfully from the original board paper — same questions, no changes.

SECTION - C (MULTIPLE CHOICE QUESTIONS)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** A

Note: Q. 15: Choose the correct answer from the given options and write the same in your answer copy. Note: Cutting and overwriting is not allowed.

Q1. Capacitance of parallel plate capacitor does not depend upon

(A) Area of plates
(B) Medium between the plates
(C) Seperation of the plates
(D) Nature of material of the plates

Q2. The absolute potential at a point distance 20 cm from a charge of 2 μc is

(A) 9×10^{2} V
(B) 9×10^{3} V
(C) 9×10^{4} V
(D) 9×10^{5} V

Q3. Which of the following represents an Electric Current?

(A) Erg c^{-1}
(B) Cs^{-1}
(C) Js^{-1}
(D) Vm^{-1}

Q4. If the resistance of 2 ohm and 4 ohm are connected in parallel, the equivalent resistance will be

(A) 6 Ohm
(B) 4 Ohm
(C) 1.5 Ohm
(D) 1.33 Ohm

Q5. Which one of the following instrument can measure the unknown resistance with sufficient accuracy?

(A) Slide wire bridge
(B) Post office box
(C) Potentiometer
(D) Galvanometer

Q6. The SI unit of magnetic induction is Tesla which is also equal to

(A) 10^{4} Gauss
(B) 10^{-4} Gauss
(C) 10^{-3} Gauss
(D) 10^{-2} Gauss

Q7. Weber is the unit of:

(A) Magnetic field intensity
(B) Magnetic induction
(C) Magnetic flux
(D) Magnetic flux density

Q8. The common doorbell requires a voltage of about

(A) 7 volts
(B) 10 volts
(C) 9 volts
(D) 4 volts

Q9. The higher frequency wave having constant amplitude and a fixed frequency is called:

(A) Modulated carrier waves
(B) Carrier waves
(C) Modulated signal
(D) Electromagnetic waves

Q10. Which one of the following substance cannot be used to form n – type substance?

(A) Aluminium
(B) Antimony
(C) Phosphorus
(D) Arsenic

Q11. The maximum speed of electrons ejected by incident light on a metal surface depends only on:

(A) Intensity of incident light
(B) Temperature of metal surface
(C) Frequency of incident light
(D) Nature of metal

Q12. The rest mass energy of an electron is:

(A) 1.02 Mev
(B) 0.511 Mev
(C) 931.5 Mev
(D) 9.11×10^{-30} Mev

Q13. Life time of meta stable state is

(A) 10^{-5} sec
(B) 10^{-4} sec
(C) 10^{-3} sec
(D) 10^{-2} sec

Q14. Mass defect per nucleon is called:

(A) Binding energy
(B) Packing fraction
(C) Average energy
(D) Excitation energy

Q15. The natural direction of flow of heat between two reservoirs depends upon

(A) Whether they are in Solids, liquid state
(B) Their pressure
(C) Their temperature difference
(D) Their heat content

Q16. The internal energy of an ideal gas depends upon

(A) Only temperature
(B) Only pressure
(C) Only volume
(D) Temperature and pressure

Q17. Various types of Cancer are treated by

(A) Cobalt – 60
(B) Stronium – 90
(C) Carbon – 14
(D) Iodine – 131

SECTION – A (Short Answer Questions)

Time Allowed: 2-40 Hours | **Maximum Marks:** 40

Note: NOTE: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. Define "Half Life" of radioactive element. The half life of $_{104}Po^{210}$ is 140 days. What would be its decay constant?

Q2. Explain the radiation technique in: (i) Food preservation (ii) Polymerization

Q3. It is possible to convert internal energy of the system into mechanical work? Explain with an example.

Q4. Two unlike capacitors of different potential and charge are joined in parallel. What happen to their potential difference? How their charges are redistributed?

Q5. Explain the combination of resistors in which the resultant resistance increases.

Q6. What do you understand by the term "LASER" summarize the laser applications?

Q7. Describe (i) Conductors (ii) Semiconductors in the light of band theory.

Q8. How does de Broglie lead to the concept of matter waves? Compute the wave length associated an electron moving at 10^7m/s?

Q9. Describe a circuit that gives a continuously varying potential difference between zero and a certain maximum value.

Q10. Define: (i) Magnetic flux (ii) Flux density along their units in SI system. Also establish relationship between them.

Q11. What is "Electron Volt"? An electron has a speed of $10^6 ms^{-1}$. Find its energy in electron volt.

SECTION – B (LONG ANSWER QUESTIONS)

Maximum Marks: 28

Note: NOTE: ATTEMPT ANY TWO (02) QUESTIONS.

Q12.

(a) [09] Explain the phenomena in which x-rays photon collide with an electron, its wave length changes. Also determine the change wave length of x – rays photon?

(b) [05] X – rays of wavelength λ_0 are scattered from carbon block at an angle of 90° with respect to the incident beam. Find the shift in wave length.

Q13.

(a) [09] Define Alternator. Explain the principle, construction and working of A.C generator. Derive an expression for motional emf developed in it.

(b) [05] A 150 turns coil in a generator has an area of 5×10^{-2} m^2 rotate in a field with B = 0.05 wbm^{-2}. How fast must the coil rotated in order to generate a maximum voltage of 150 volts?

Q14.

(a) [09] State and explain the statements which are based on the working of heat engine and refrigerator. Show that both the statements are equivalent.

(b) [05] Determine the average value of the Kinetic energy of particles of an ideal gas at 348K.